

Kazi Ahsan Ahmed Akaid

Mechanical Engineering | Machine Learning | Prognostics & Health Management
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RESEARCH INTERESTS

Prognostics and Health Management (PHM) | Remaining Useful Life (RUL) Prediction | Explainable AI | Physics-Informed Machine Learning
| Intelligent Manufacturing | Industrial Reliability | Predictive Maintenance

EDUCATION

Chittagong University of Engineering & Technology (CUET)

Chattogram, Bangladesh

Bachelor of Science in Mechanical Engineering

Feb. 2022 – Jul. 2026

CGPA: 3.55/4.00 | Final 4-semester average: 3.80/4.00

Thesis: Predictive Modeling of Nanocellulose Characteristics from Acid Hydrolysis Process Parameters Using Machine Learning

RESEARCH EXPERIENCE

Research Intern — ELITE Research Lab

Sep. 2026 – Present

ELITE Research Lab

Undergraduate Thesis Researcher

2025 – 2026

Department of Mechanical Engineering, CUET

Chattogram, Bangladesh

- Compiled a **341-record literature dataset** linking cellulose source and acid-hydrolysis parameters—acid concentration, temperature, and reaction time—to cellulose nanocrystal (CNC) length and crystallinity.
- Developed and compared **seven regression models** using hyperparameter tuning and cross-validation, with paper-level grouping to reduce data leakage and assess generalization across literature sources.
- Applied **SHAP and impurity-based feature-importance analysis** to interpret model predictions and evaluated **90% conformal prediction intervals** for uncertainty quantification.

Independent Researcher — Prognostics & Explainable AI

2025 – Present

Independent Research

Chattogram, Bangladesh

- Designed a **stacking ensemble** combining Random Forest, XGBoost, LightGBM, and CatBoost with a Ridge meta-learner for Remaining Useful Life prediction using the **NASA C-MAPSS FD001 and FD003 datasets**.
- Developed a predictive modeling pipeline covering sensor-data preprocessing, feature engineering, ensemble learning, model evaluation, and explainability analysis.
- Used **Group K-Fold cross-validation** to evaluate model generalization and assessed performance using RMSE and the **NASA PHM08 scoring function**.
- Introduced quantitative **cross-model explanation agreement metrics**, including Feature Agreement, Rank Agreement, Pairwise Rank Agreement, and Spearman correlation, using SHAP TreeExplainer.
- Performed **leakage-free SHAP-based feature pruning** to investigate the trade-off between predictive accuracy, feature dimensionality, and inference efficiency.
- Developing a research framework on **uncertainty-quantified explainable deep learning** for tool wear prediction in high-speed CNC milling, integrating predictive modeling, uncertainty estimation, and explainability for intelligent manufacturing.

PUBLICATIONS & MANUSCRIPTS

Conference Papers

- **Akaid, K. A. A.**, and Rabbi, S. “Predictive Modeling of Nanocellulose Characteristics from Acid Hydrolysis Process Parameters Using Machine Learning.” *Submitted*.
- **Akaid, K. A. A.**, et al. “Turbofan Engine Remaining Useful Life Prediction: Stacking Ensembles, Explanation Agreement, and Feature Pruning.” *Submitted to ICMIEE 2026*, Khulna University of Engineering & Technology.

Journal Manuscript

- **Akaid, K. A. A.**, et al. “Uncertainty-Quantified Explainable Deep Learning for Tool Wear Prediction in High-Speed CNC Milling.” *Manuscript in preparation*.

SELECTED RESEARCH PROJECTS

Turbofan Engine RUL Prediction — NASA C-MAPSS

2025 – 2026

- Built an end-to-end pipeline from raw multivariate sensor data through preprocessing, feature engineering, ensemble learning, model evaluation, and explainability analysis.
- Investigated explanation consistency across heterogeneous machine learning models as an additional perspective for assessing model reliability in PHM applications.
- Code: github.com/aakaid010/RUL-Prediction-using-Nasa-C-MAPSS-DATA

NanoCNC Predictor

2025 – 2026

- Developed a Flask-based machine-learning application for predicting nanocellulose length and crystallinity from acid-hydrolysis process parameters.
- Code: github.com/aakaid010/NanoCNC-Predictor

INDUSTRIAL EXPERIENCE

Industrial Attachment — Bangladesh Industrial Technical Assistance Center (BITAC) Chattogram, Bangladesh

- Completed an industrial attachment at BITAC, gaining practical exposure to manufacturing processes, machine tools, and industrial engineering practices.
- Observed conventional and non-conventional machining processes, CNC machinery, welding and fabrication, heat treatment, foundry and pattern making, machine maintenance, and inspection and testing.
- Gained practical exposure to the operation and application of manufacturing and machining equipment in an industrial training environment.

Industrial Visit — S. Alam Cold Rolled Steels Limited Chattogram, Bangladesh

- Visited S. Alam Cold Rolled Steels Limited to observe industrial steel processing and manufacturing operations.
- Gained first-hand exposure to cold-rolled steel production and industrial manufacturing practices.

TECHNICAL SKILLS

- **Programming:** Python, MATLAB
- **Machine Learning:** scikit-learn, Random Forest, Gradient Boosting, SVR, XGBoost, LightGBM, CatBoost, Stacking Ensembles
- **Explainable AI:** SHAP, Grad-CAM, Feature Importance Analysis, Cross-Model Explanation Agreement Metrics
- **Data Science:** NumPy, pandas, Matplotlib, Feature Engineering, Feature Selection, Cross-Validation, Hyperparameter Optimization, Model Evaluation
- **PHM / RUL:** NASA C-MAPSS, IMS, Paderborn, RUL Prediction, Predictive Maintenance
- **Engineering / Software:** SolidWorks, Flask, Git/GitHub, LaTeX
- **Research Platforms:** Google Colab, Kaggle

RELEVANT COURSEWORK

Numerical Methods in Engineering | Computer Programming | MATLAB | Statistical Analysis | Measurement & Quality Control | Mechanics of Machinery & Vibration | Fluid Mechanics I & II | Engineering Metallurgy

AWARDS & LEADERSHIP

- **Dean's List**, Department of Mechanical Engineering, CUET | 3 semesters
- **Organizing Secretary**, Robo Mechatronics Association (RMA), CUET
 - Coordinated student events and activities for the university's robotics and mechatronics association.
- **General Member**, ASME Student Chapter, CUET
 - Participated as a general member of the ASME CUET student chapter.

REFERENCES

1. Dr. Md. Sanaul Rabbi

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Chittagong University of Engineering & Technology

Relationship: Academic Advisor

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2. Redoy Masum Meraz

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Chittagong University of Engineering & Technology

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